

RUIQI HE

CONTACT

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EDUCATION

The Ohio State University|Columbus, OH

Bachelor of Science, Data Analytics | Specialization: Biomedical and Public Health Analytics | Minor: Business

ASC Honors College | Dean's List | Expected Graduation: May 2028

SKILLS

- Programming Languages: Python, SQL, R, Java, C++, TypeScript/JavaScript
- Machine Learning & AI: PyTorch, scikit-learn, Pandas, prompt engineering, LLMs (Gemini, Qwen, IBM watsonx), NVIDIA RAPIDS cuDF
- Data Engineering & Analytics: Databricks, geospatial analysis, large-scale preprocessing, Monte Carlo / VaR-ES
- Software & Tools: FastAPI, React, Node.js/Express, Docker, Git/GitHub Actions, NVIDIA DGX
- Spoken Languages: English (Native), Chinese (Native), Spanish (Basic)

EXPERIENCE

Undergraduate Research Intern – AIMed Lab | Summer 2026 | The Ohio State University | Department of Biomedical

Mentor: Dr. Ping Zhang

- Competing in the 2026 George B. Moody PhysioNet Challenge to develop predictive models that forecast cognitive impairment diagnoses from polysomnography (sleep study) data
- Engineering ML pipelines to analyze multi-modal physiological signals including EEG, EOG, EMG, ECG, respiratory effort, and airflow to identify patients at risk for mild cognitive impairment, Alzheimer's disease, and dementia
- Processing large-scale PSG datasets (1.2TB training set across 6,600+ recordings) with heterogeneous channel availability, variable sampling rates, and real-world noise
- Optimizing age-conditioned AUROC scoring metrics to predict diagnoses occurring 1-6 years post-sleep study

PROJECTS

ASA DataFest @ OSU | Best Visualization Award | Spring 2026

Competition Project for Stormont Vail Health

- Engineered a dynamic geospatial dashboard with regional health trend identification, lasso-based multi-region selection, department-level filters, and top-diagnosis frequency analysis
- Leveraged NVIDIA RAPIDS cuDF on the DGX Spark to accelerate data preprocessing and cleaning, significantly improving dashboard responsiveness for real-time filtering across 100K+ patient encounters
- Extracted actionable insights for hospital administration on patient distribution patterns and healthcare access barriers across Kansas, enabling data-driven resource allocation decisions

CLARA (Clairvoyant Loss Avoidance & Risk Advisor) | Hackathon Win | Spring 2026

- Built a prototype AI-driven enterprise risk management platform for investors and portfolio managers
- Implemented real-time VaR & Expected Shortfall monitoring, Monte Carlo simulations, and hedge optimization concepts
- Automated 10-K risk extraction, regime detection, and stress testing for demonstration
- Leveraged Gemini for financial reasoning and Watson AI for conversational insights

NYC Housing Sales Analytics Website | Hackathon Winner | Spring 2026

- Built a full-stack analytics platform analyzing 34,394 NYC property sales across all five boroughs using Python and Databricks
- Developed an end-to-end data pipeline for cleaning, geospatial enrichment, feature engineering, and ML-based price prediction
- Created an interactive dashboard for market trends, borough comparisons, property insights, and fair-market value predictions

Doctor-in-the-Loop AI System | 1st Place — Foundation of AI Track | Autumn 2025

- Developed an explainable X-ray AI workflow combining model findings with visual overlays for interpretable clinical analysis.
- Implemented editable AI-generated treatment recommendations, allowing clinicians to review and modify proposed plans.
- Designed the system around human-in-the-loop decision-making, keeping final clinical decisions under physician control.

Talkora — Gemini Hackathon | Spring 2026

- Built a real-time, voice-first AI language tutor using the Gemini Live API, supporting adaptive conversations across different levels
- Implemented AI personas with live pronunciation and grammar feedback designed to maintain natural conversational flow.
- Engineered a low-latency browser-to-AI audio pipeline using the Web Audio API and Google AI Studio.

AI Study Planner | Top Project — BDAA x Lovable Hackathon | Autumn 2025

- Built an AI-powered platform that transforms notes, documents, and lecture recordings into personalized study schedules and exam-prep plans.
- Implemented AI summarization, adaptive scheduling, and auto-generated quizzes to reinforce learning and track student progress.

ACTIVITIES

- BDAA(Big Data & Analytics Association): Autumn 2025 - Present
 - Foundation of AI TA: Spring 2026 - Present
- Formula Buckeye: Autumn 2025 - Present
- Academic Team: Autumn 2022 – Spring 2024 (PA State Academic Team Champion)
- Mock Trial: Autumn 2022 – Spring 2024